

L10a: Sensitivity Analysis of American Options Premium

CHEME 5660 Cornell University

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Objectives for Today

In L2b, duration and convexity told us how much a bond's price changes when its yield moves. In L9b, we priced American options on a CRR tree. How much does an option's premium change when the share price, volatility, time to expiration, or interest rate changes?

Today's concepts

- Compute the Greeks for American options: Delta, Theta, Vega, Rho, and Gamma from a local Taylor expansion of the options premium.
- Interpret Delta as a hedge ratio and a probability estimate: the shares needed to hedge an option position, compared with the probability of expiring in-the-money.
- Calculate probability of profit and probability of exercise: from projected share price distributions, at the breakeven and at the strike price.

Lectures and Examples

Lecture:
CHEME-5660-L10a-Lecture-American-Derivatives-Greeks-Fall-2026.ipynb

Example 1:
CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

Example 2:
CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

The first example computes the Greeks for call and put options from market data. The second computes the probability of profit and the probability of exercise for a short call, and compares them with Delta.

Concept Review: Intrinsic and Extrinsic Value

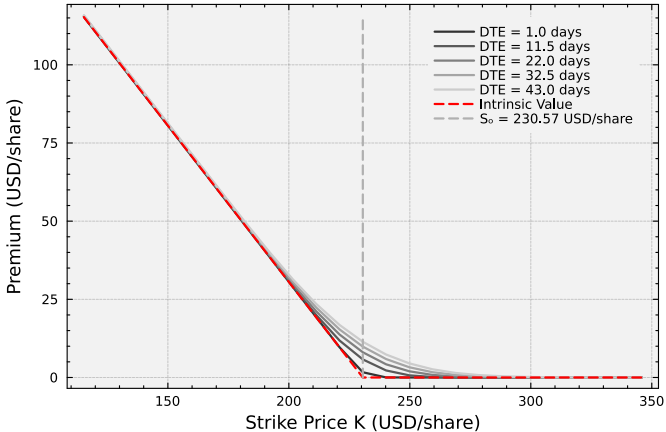
In the last lecture, we split the American premium into two components:

- Intrinsic value: the immediate exercise value, $\max(0, S_t - K)$ for a call and $\max(0, K - S_t)$ for a put.
- Extrinsic value: also known as time value, the difference between the option price and its intrinsic value.

$$E_V(t) = \begin{cases} \mathcal{P}_c - \max(0, S_t - K) & \text{(for Call Options)} \\ \mathcal{P}_p - \max(0, K - S_t) & \text{(for Put Options)} \end{cases}$$

When the option is out of the money, it has no intrinsic value, and the entire option price is extrinsic value.

Call Premium across Strikes and Expirations



Hmmmm: the premium changes with time to expiration and the strike price. How does it change with other factors?

Company Profile: Susquehanna

Susquehanna (SIG) is a privately held trading firm whose core business is options trading. Friends who traded options on the floor of the Philadelphia Stock Exchange started it in 1987.

- Options market making: it posts prices to buy and sell, so it takes the other side of its customers' contracts. In October 2026, it was the Designated Primary Market Maker for 1,359 of 5,370 Cboe option classes, the most of any firm.
- Managing the Greeks: its book is exposed to Delta, Vega, Theta, and Rho at once, and the goal is to net these across products.
- Training: interns learn options theory, and poker teaches new traders to decide under uncertainty.

Connection to today's lecture: when a customer buys a call, the firm hedges with $100\,\Delta_c$ shares and rebalances as Gamma moves Δ_c.

The Greeks

The Greeks are local sensitivity coefficients that measure how American options premiums change with market conditions:

- Delta Δ: the change in premium for a `+1 USD/share` change in the share price.
- Theta Θ: the change in premium for a `1-day decrease` in the lifetime of the contract (time decay).
- Vega ν: the change in premium for a `+1%` change in volatility.
- Gamma Γ: the change in Δ for a `+1 USD/share` change in the share price.
- Rho ρ: the change in premium for a `+1%` change in the risk-free rate, usually minor except for longer-term options.

Most trading platforms compute these for you!

A Local Taylor Expansion of the Premium

Expanding the premium $\mathcal{P}_\star(S, \sigma, t, g_y)$ of a contract of type $\star \in \{c, p\}$ to second order in S and first order in the other inputs:

$$\boxed{\Delta\mathcal{P}_\star \approx \Delta_\star\,\Delta S + \nu_\star\,\Delta\sigma + \Theta_\star\,\Delta t + \rho_\star\,\Delta g_y + \tfrac{1}{2}\,\Gamma_\star\,(\Delta S)^2}$$

$$\Delta_\star = \frac{\partial \mathcal{P}_\star}{\partial S} \qquad \nu_\star = \frac{\partial \mathcal{P}_\star}{\partial \sigma} \qquad \Theta_\star = \frac{\partial \mathcal{P}_\star}{\partial t} \qquad \rho_\star = \frac{\partial \mathcal{P}_\star}{\partial g_y} \qquad \Gamma_\star = \frac{\partial^2 \mathcal{P}_\star}{\partial S^2}$$

Derivatives are taken at the current inputs, with Δt in years and $\Delta\sigma$, Δg_y in decimals. Platforms quote $\Theta_\star/365$ per day, and $\nu_\star/100$ and $\rho_\star/100$ per percentage point.

Delta and Gamma as Duration and Convexity

$\Delta_\star$ and $\Gamma_\star$ play the roles of duration and convexity in L2b: the slope and the curvature of the premium as a function of the share price.

Example: a long call has $\Delta_c = 0.50$ and $\Gamma_c = 0.02$ per USD/share.

$$\Delta S = +5: \quad \Delta \mathcal{P}_c \approx 0.50 \times 5 + \tfrac{1}{2}(0.02)(5)^2 = 2.50 + 0.25 = 2.75 \text{ USD/share}$$

$$\Delta S = -5: \quad \Delta \mathcal{P}_c \approx -2.50 + 0.25 = -2.25 \text{ USD/share}$$

The Gamma term adds to the premium whichever way the share price moves.

Computing the Greeks of American Options

Backward induction on the CRR tree gives no formula to differentiate, so we reprice the tree with one input moved up and down by a step ϵ:

$$\frac{\partial \mathcal{P}_\star}{\partial x} \approx \frac{\mathcal{P}_\star(x+\epsilon) - \mathcal{P}_\star(x-\epsilon)}{2\epsilon} \qquad x \in \{S, \sigma, t, g_y\}$$

$$\Gamma_\star \approx \frac{\mathcal{P}_\star(S+\epsilon) - 2\,\mathcal{P}_\star(S) + \mathcal{P}_\star(S-\epsilon)}{\epsilon^2}$$

With no dividends, a call's American and European premiums are equal (L9b), so the L9a Black–Scholes–Merton formulas give Greeks to check against, e.g., $\Delta_c = N(d_+)$.

Example 1: CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

The Secret Lives of Delta

Besides measuring the premium change for a `+1 USD/share` move in the share price, Δ has two other interpretations:

- Hedge ratio: for every n contracts held ($n < 0$ if the contracts are sold), hold $-100 \cdot n \cdot \Delta$ shares to be delta-neutral, i.e., insensitive to small share price moves.
- Probability of exercise: Δ is often used as an estimate of the probability that the option expires in-the-money. A call with $\Delta = 0.7$ suggests a 70% chance.

Probability of Exercise (POE)

The probability that the contract finishes in-the-money at expiration $t = T$:

$$\text{POE} = \begin{cases} \mathbb{P}\left(S_T > K + 0.01\right) & \text{(for Call Options)} \\ \mathbb{P}\left(S_T < K - 0.01\right) & \text{(for Put Options)} \end{cases}$$

In the L9a Black–Scholes–Merton model, a European call has $\Delta_c = N(d_+)$, while the risk-neutral probability that it finishes in-the-money is $N(d_-)$, which is smaller.

Conjecture: $\Delta \approx$ POE. Example 2 checks it.

Probability of Profit (POP)

The probability that the contract's profit at expiration is positive:

$$\boxed{\text{POP} = \mathbb{P}\left(\hat{P}_{\mathcal{C}} > 0\right)}$$

where $\hat{P}_{\mathcal{C}}$ is the profit of the (single or composite) contract at expiration.

POE reads the projected distribution at the strike price, POP at the breakeven.

Example 2: CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Optional Advanced Material

Numerical Greeks: advanced/numerical_greeks/
CHEME-5660-L10a-Advanced-Numerical-Greeks-Fall-2026.ipynb

How accurate are Greeks computed by repricing a binomial tree? The CRR premium bends only where a terminal node crosses the strike, so a step that stays between two bends gives a Gamma of zero. The Leisen–Reimer tree keeps the call Greeks close to the Black–Scholes–Merton values at every number of tree steps we try. We then plot an American put's Delta and Gamma around the early-exercise boundary.

Notebooks: lectures/week-10/L10a/advanced, also linked from the lecture notebook.

Summary

Today we introduced the Greeks, which measure how American options premiums respond to changes in market conditions.

Key takeaways

- The Greeks quantify option price sensitivity: they are the partial derivatives in a local Taylor expansion of the premium, which we estimate for American options by repricing the CRR tree.
- Delta serves three purposes: the rate of change in the premium, the hedge ratio for delta-neutral positions, and an approximation of the probability of exercise.
- Probability of profit differs from probability of exercise: POE reads the projected distribution at the strike price, and POP reads it at the breakeven.